

Teaching Exam

Subject – Computer Science

Topic – Computer Network



Lecture No.- 09

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Topics to be Covered

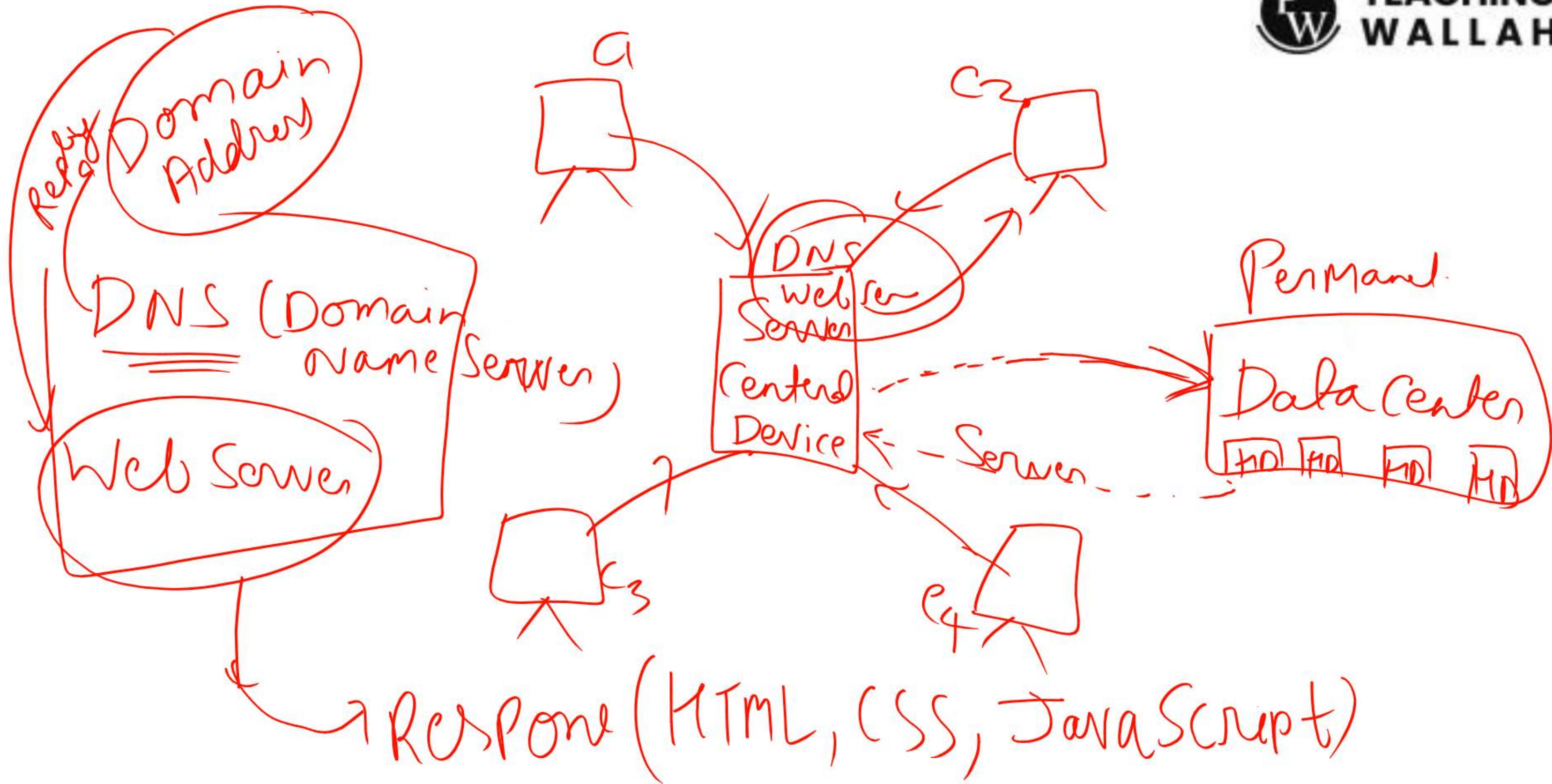
Topic



- 1:- SMTP PROTOCOL AND ITS WORKING MODEL**
- 2:- POP3 PROTOCOL**
- 3:- IMAP PROTOCOL**
- 4:- Difference between POP3 and IMAP Protocol**

* Client → Client is Machine which seeking the Service from Server
* Client are More than One

* Server → Server is Machine which seeking the Request from different Client Machine
* Server Machine is always One in Quantity



SMTP represents **Simple Mail Transfer Protocol**. SMTP is a set of interaction guidelines that allow the software to transmit electronic mail over the internet, referred to as Simple Mail Transfer Protocol.

The main objective of SMTP is used to set up communication rules between servers. The servers have a way of recognizing themselves and announcing what kind of communication they are trying to perform. They also have a way of handling errors such as incorrect email address. For example, if the recipient address is wrong, then receiving a server reply with an error message of some kind.

To send an email, the client opens a TCP connection to the SMTP server. The server, which is always listening on port 25, initiates the connection as soon as it detects a client. Once the TCP connection is established, the client sends the email across the connection.

SMTP

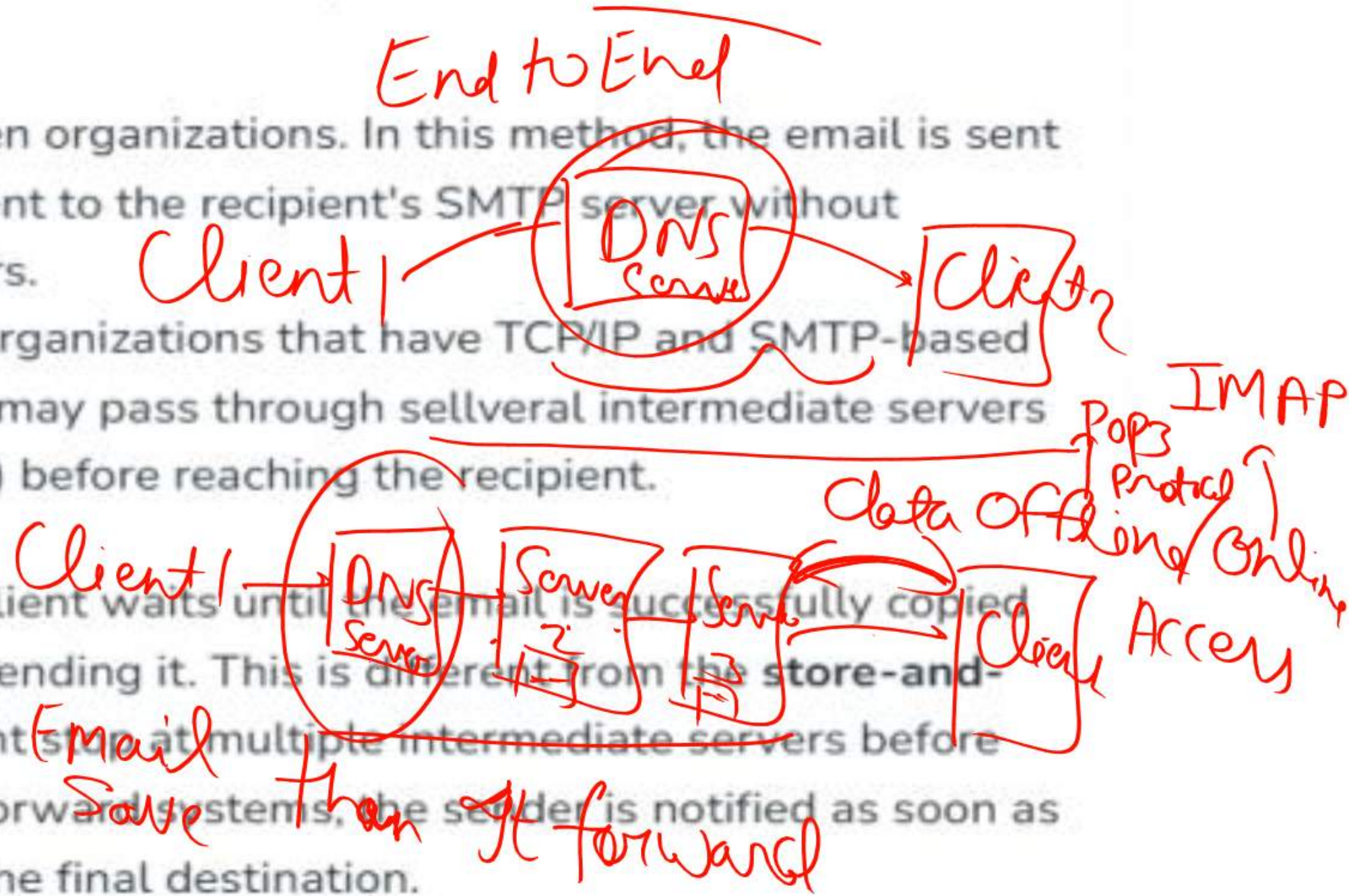


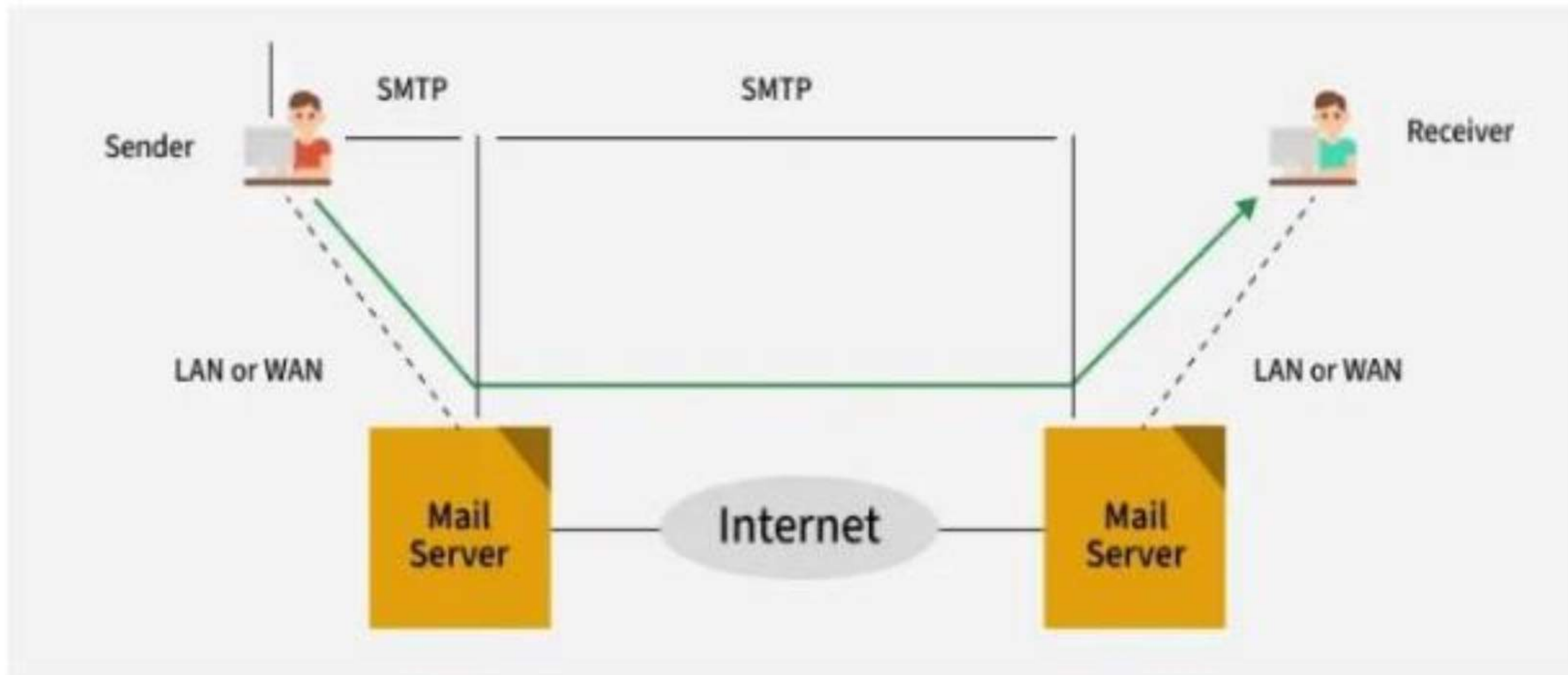
Types of SMTP Protocol

The SMTP model supports two types of email delivery methods: **end-to-end** and **store-and-forward**.

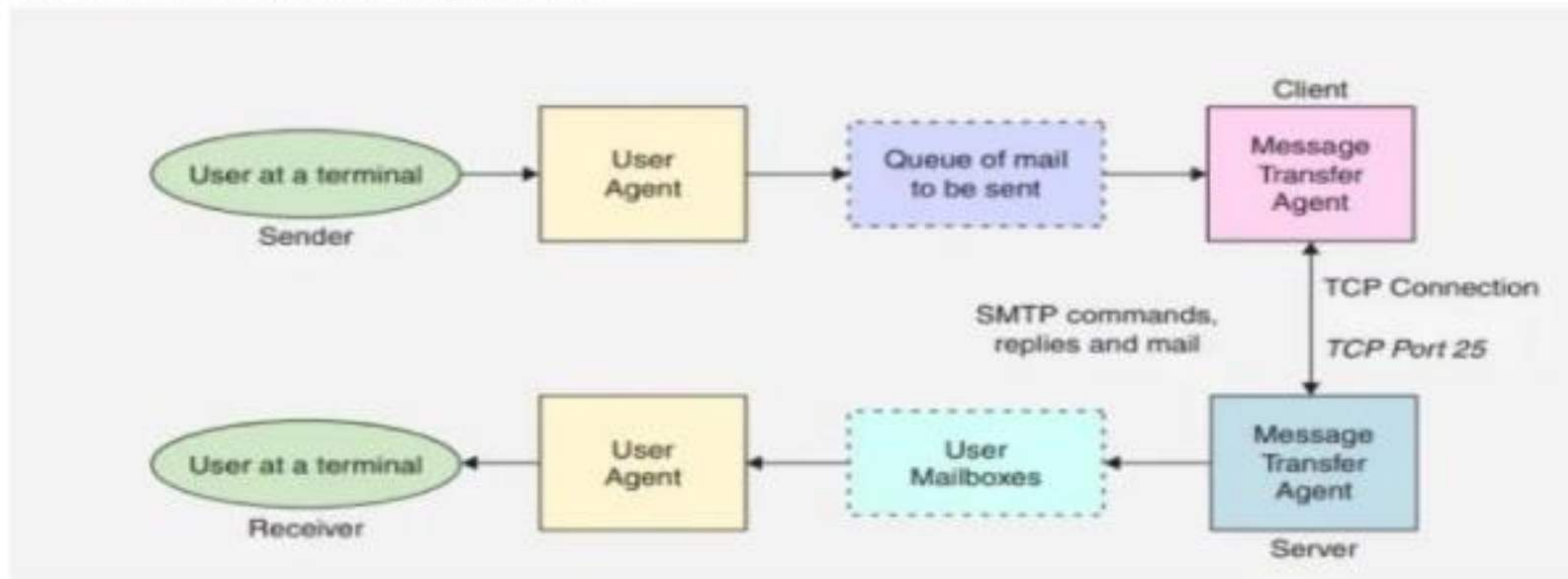
- **End-to-end** delivery is used between organizations. In this method, the email is sent directly from the sender's SMTP client to the recipient's SMTP server without passing through intermediate servers.
- **Store-and-forward** is used within organizations that have TCP/IP and SMTP-based networks. In this method, the email may pass through several intermediate servers (Message Transfer Agents, or MTAs) before reaching the recipient.

With **end-to-end** delivery, the SMTP client waits until the email is successfully copied to the recipient's SMTP server before sending it. This is different from the **store-and-forward** method, where the email might stop at multiple intermediate servers before reaching its destination. In store-and-forward systems, the sender is notified as soon as the email reaches the first server, not the final destination.





Model of SMTP System



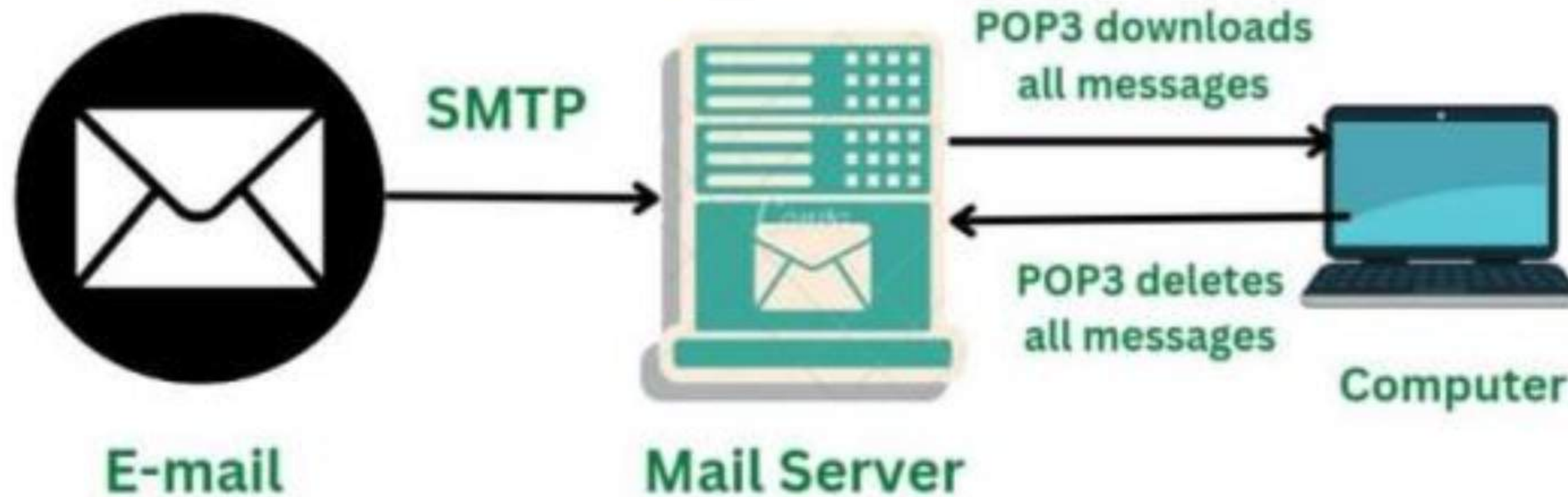
Introduction to POP

POP stands for Post Office Protocol. The [POP protocol](#) was published in the year 1984. POP has been updated two times namely "POP2" and "POP3". The POP protocol is an Internet Standard Protocol that works on the application layer. It is used to get an access email from the mail server. The need for POP mainly arises when the user or client does not have a continuous internet connection and wants to receive email messages. The Pop client makes use of POP to pull email messages from the POP server. POP3 is the updated version of POP.

What is POP3?

POP 3 stands for Post Office Protocol Version 3. POP3 protocol is used to provide access to the mail inbox that is stored in the email server. POP3 protocol can download and delete messages. Once the POP3 client has established a connection with the mail server it can easily retrieve all the messages from the server. The user can access the messages locally even if the user is offline. The protocol does not inherently support real-time synchronization or automatic checking for new messages, users can configure their email clients to check for new messages at intervals or manually. Many email programs, including Apple Mail, Gmail, and Microsoft Outlook, support the POP3 protocol, although IMAP is often preferred for its synchronization features.

When a message is sent, SMTP is used to transfer it from the client to the server and ultimately to the server of the recipient. However, the Message Access Agent facilitates the transmission of the message from the receiving server to the host server. POP3 and IMAP are the two types of protocols that are included in the Message Access Agent.

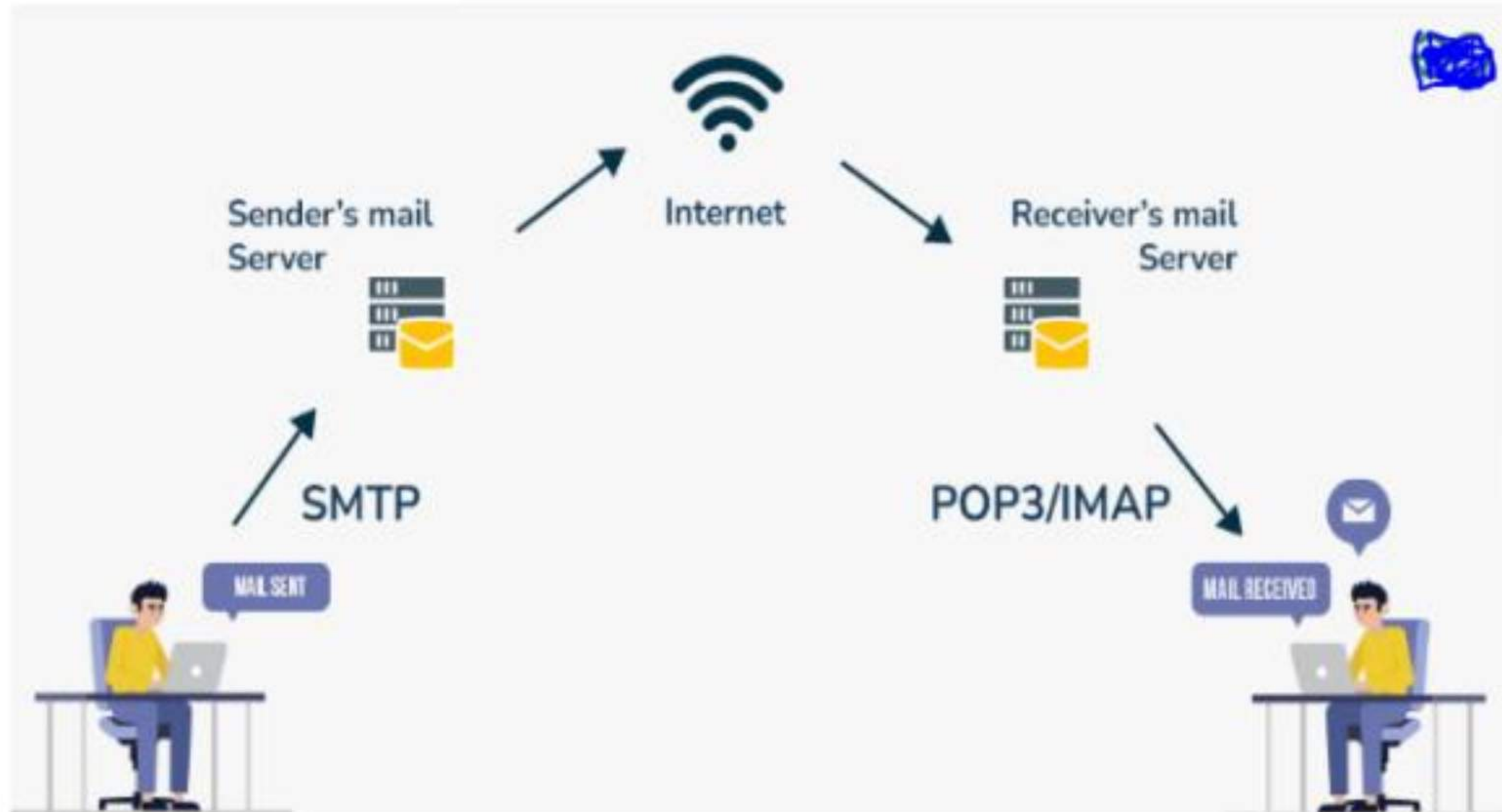


Advantages of POP3

- The message can be read offline also.
- POP3 requires less storage space.
- POP3 is easy to use and configure.
- POP3 is supported by many email applications.
- Since our PC already has the emails stored there, accessing them is quick and simple.
- The size of emails we send and receive is unrestricted.
- Since all emails are kept locally, less server storage space is needed.

Disadvantages of POP3

- POP3 does not provide the feature of real-time synchronization.
- If an email consists of virus attachments it can affect the system easily.
- POP3 does not support accessing the same email at the same time on different systems.
- At a time entire email folder can be discarded.
- It is also possible for the email folder that is downloaded from the mail server to get



Internet Message Access Protocol (IMAP) is an application layer protocol that operates as a contract for receiving emails from the mail server. It was designed by Mark Crispin in 1986 as a remote access mailbox protocol, the current version of [IMAP](#) is IMAP4.

It is used as the most commonly used protocol for retrieving emails. This term is also known as Internet mail access protocol, Interactive mail access protocol, and Interim mail access protocol.

It retrieves messages from your email provider to your email client (such as on your Mac or PC). Importantly, it does not delete messages from the email service after you download them until the user explicitly deletes them.

It allows you to view and manage your emails across multiple devices because it synchronizes the emails between your email client and the server. This synchronization ensures that changes such as read status, deletions, and folder organization are reflected consistently across all devices connected to the same email account.

Working of IMAP

IMAP follows Client-server Architecture and is the most commonly used email protocol. It is a combination of client and server process running on other computers that are connected through a network.

This protocol resides over the TCP/IP protocol for communication. Once the communication is set up the server listens on port 143 by default which is non-encrypted. For the secure encrypted communication port, 993 is used.

- **Port 143:** Default non encrypted IMAP communication.
- **Port 993:** Secure IMAP communication with encryption (SSL/TLS).

Advantages

- It offers synchronization across all the maintained sessions by the user.
- It provides security over [POP3 protocol](#) as the email only exists on the IMAP server.
- Users have remote access to all the contents.
- It offers easy migration between the devices as it is synchronized by a centralized server.
- There is no need to physically allocate any storage to save contents.

Disadvantages

- IMAP is complex to maintain.
- Emails of the user are only available when there is an internet connection.
- It is slower to load messages.
- Some emails don't support IMAP which makes it difficult to manage.

Post Office Protocol (POP3)	Internet Message Access Protocol (IMAP)
<u>POP</u> is a simple protocol that only allows downloading messages from your Inbox to your local computer.	<u>IMAP (Internet Message Access Protocol)</u> is much more advanced and allows the user to see all the folders on the mail server.
The POP server listens on port 110, and the POP with SSL secure(POP3DS) server listens on port 995	The IMAP server listens on port 143, and the IMAP with SSL secure(IMAPDS) server listens on port 993.
In POP3 the mail can only be accessed from a single device at a time.	Messages can be accessed across multiple devices
To read the mail it has to be downloaded on the local system.	The mail content can be read partially before downloading.

Topic- Difference between POP3 and IMAP Protocol



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In POP3 the mail can only be accessed from a single device at a time.	Messages can be accessed across multiple devices
To read the mail it has to be downloaded on the local system.	The mail content can be read partially before downloading.
The user can not organize mail in the mailbox of the mail server.	On the mail server, the user can directly arrange the email.
The user can not create, delete, or rename email on the mail server.	The user can create, delete, or rename an email on the mail server.
It is unidirectional i.e. all the changes made on a device do not affect the content present on the server.	It is Bi-directional i.e. all the changes made on the server or device are made on the other side too.
It does not allow a user to sync emails.	It allows a user to sync their emails.

Changes in the mail can be done using local email software.	Changes made to the web interface or email software stay in sync with the server.
All the messages are downloaded at once.	The Message header can be viewed before downloading.

THANK YOU